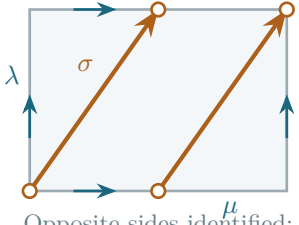
Primitive cusp marking

Schematic torus fundamental domain



Opposite sides identified;
amber segments join into one loop.

$\sigma = \mu + p\lambda \qquad$ (drawn: $p = 2$)

$U + pV = -2\pi i$

Peripheral square-root cover

Local sheets of the μ_2 torsor $\ell^2 = L$

selected branch
(M, ℓ)
$\ell = e^{V/2} \longrightarrow 1$

$(M, -\ell)$
other local sheet

$(M, L) \in (\mathbb{C}^\times)^2$

$\pi(M, \ell) = (M, \ell^2)$

$(M^0, \ell) \longmapsto (M^0 \ell^{-\varepsilon}, \ell)$

Integral exponents on the cover; $\varepsilon \in \{-1, 0, 1\}$.

$G_\varepsilon = G_0 - \varepsilon V^2/8$

Local peripheral coordinates and action change.
Each filled topology retains its own primitive meridian and longitude.